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(See 6.8)

INTERIM FEDERAL SPECIFICATION

SANDERS, ELECTRIC, PORTABLE, BELT, DISK AND ORBITAL PAD, DOUBLE INSULATED

This Interim Federal Specification was developed by the Department of the Navy, Naval Ship Engineering Center, Center Building, Prince George's Center, Hyattsville, Maryland 20782, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal Specification pending the issuance of a coordinated document.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers double insulated, portable electric, belt, disk and orbital pad sanders.

1.1.1 Federal specification coverage. Federal specifications do not include all the types, classes, styles, and sizes of the commodity indicated by the title of this specification, but are intended to cover those most generally used by the Federal Government.

1.2 Classification. Double insulated, portable electric sanders covered by this specification shall be furnished in the following types, styles, and sizes, as specified (see 6.2):

Type I - Sanders, belt.

Style 1 - Without dust collecting system.
Style 2 - With dust collecting system.
Style 3 - With connector for attachment to external dust collecting system.

Size 1 - 3 inch wide by 21 inch long belt.
Size 2 - 3 inch wide by 24 inch long belt.
Size 3 - 4 inch wide by 27 inch long belt.

Type II - Sanders, disk.

Style 1 - Straight handle.

Size 7 - 7 inch diameter abrasive disk.
Size 9 - 9 inch diameter abrasive disk.

Style 2 - Side handle.

Type III - Sanders, orbital pad.

Style 1 - One hand grip.
Style 2 - Two hand grip.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:
PPP-P-40 - Packaging and Packing of Hand Tools.

FSC 5130

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Federal Standard:
Fed. Test Method Std. No. 406 - Plastics: Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo, Fort Worth, Denver, Los Angeles, San Francisco, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:
MIL-V-173 - Varnish, Moisture and Fungus-Resistant (For the Treatment of Communications, Electronic and Associated Electrical Equipment).

Military Standards:
MIL-STD-102 - Anti-Friction Bearing Identification Code.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1337 - General Suppression System Design Requirements For Portable Electric Hand Tools.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H-28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

UNDERWRITERS' LABORATORIES, INC.
UL 45 - Electric Tools, Standards for Safety.
UL 498 - Attachment Plugs and Receptacles.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 333 Pfingsten Road, Northbrook, Illinois 60062; 1285 Walt Whitman Road, Melville, Long Island, New York 11746; and 1655 Scott Boulevard, Santa Clara, California 95050.)

American National Standards Institute (ANSI) Standards
B5.38 - Driving and Spindle Ends For Portable Air and Electric Tools.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

The Anti-Friction Bearing Manufacturers' Association (AFBMA) Standards
Section 1 - Terminology for Anti-Friction Ball and Roller Bearings and Parts.
Section 2 - Boundary Dimensions for Ball and Roller Bearings.
Section 3 - Ball and Roller Bearings - Bearing Tolerances.
Section 5 - Bearing Identification Code for Anti-Friction Ball and Roller Bearings.
Section 7 - Bearing Mounting for Ball and Roller Bearings.

(Application for copies should be addressed to the Anti-Friction Bearing Manufacturers' Association Inc., 60 East 42nd Street, New York, New York 10017.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Double insulated portable electric sanders furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.5).

3.2 Illustrations. The illustrations shown herein are descriptive and not restrictive and are not intended to preclude the purchase of sanders which are otherwise in accordance with this specification.

3.3 Material. All materials used in the fabrication of double insulated electric sanders shall be new and of the quality necessary to produce sanders to meet the requirements described herein. Electrical materials and components shall conform to the standards of the Underwriters' Laboratories (UL), Inc.

3.4 Design. The sanders furnished shall be of the manufacturer's current design and shall include industry's latest developments in the double insulated, tool field. The sanders shall be of the standard commercial heavy duty portable electric sander design. All parts subject to wear shall be easily accessible for adjustment, replacement, and repair.

3.4.1 Safety. Electric sanders shall be designed for ease of operation and safety in use. All parts which are hazardous to the operator shall be suitably guarded against accidental contact where such guards do not interfere with direct operation of the tool. The enclosure, except for the ventilating openings, shall totally enclose all stationary and moving parts and provide adequate protection against accidental contact with live parts even after the removal of detachable parts. Ventilating slots shall conform to the requirements for accessibility of live parts of UL 45. Creepage or clearance distance through ventilating air slots from live parts to metal foil wrapped around the exterior of the case shall not be less than 3/16 inch. In addition to protection from moving parts, the sanders shall be so designed and constructed that in the event of defects which may occur in the functional insulation, there shall be no danger of shock to the operator when tested in accordance with UL 45 and Section 4 herein.

3.4.1.1 Conformance to standards of Underwriters' Laboratories Inc. The contractor shall submit to the contracting agency proof that the basic sander he proposes to supply under this specification conforms to the requirements of the Underwriters' Laboratories, Inc. Standards for Electric Tools (UL 45). The label, or listing with re-examination of the UL may be accepted as evidence that the sander conforms to this requirement. In lieu of the label, or listing with re-examination, the contractor may submit independent proof, satisfactory to the contracting agency, that the sander conforms to the applicable requirements of the published standards including methods of tests of the UL Standards noted above. Compliance with requirements of UL regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his sander. In event of difference between any of the requirements of this specification and those of UL, the requirements of this specification shall govern.

3.4.2 Insulation. Each tool shall be provided with double insulation as defined (see 6.6). The purpose of secondary protective insulation shall be to provide shock hazard protection in the event of primary insulation failure. Sanders shall conform to the insulation requirements of UL 45 except as specified herein.

3.4.2.1 Creepage and clearance distances. Creepage and clearance distances shall comply with the spacings specified in Table I.

3.4.3 Lubrication. Bearings, except prelubricated sealed bearings, gears and moving parts shall be assembled with lubricating medium. Convenient access shall be provided to area where manufacturer recommends periodic lubrication or lubricant replenishment is required. External lubricant fittings shall not be provided. Lubricant shall not leak from gear casing or enter motor compartment. Parts requiring protection from dust shall be enclosed.

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Table I - Spacings.

Parts between which spacings are measured	Acceptable spacings
1. Between live parts and dead metal parts separated by functional insulation.	3/32 inch along insulating surfaces and through air.
2. Between accessible dead metal parts and dead metal parts separated from live parts by functional insulation.	3/32 inch along insulating surfaces and through air.
3. Between exterior of the housing (metal foil) and dead metal parts separated from live parts by functional insulation.	3/32 inch along insulating surfaces and through air.
4. Between live parts and accessible dead metal parts separated by reinforced insulation.	3/16 inch along insulating surfaces and through air.
5. Between live parts and exterior of the housing (metal foil) of insulating material.	3/16 inch along insulating surfaces and through air.

3.4.4 Interchangeability. All parts from the same manufacturer shall be manufactured to such standards as will permit interchangeability of similar parts without modification or adjustment.

3.5 Construction. Construction shall be free of any characteristics or defects that will prevent sanders from passing any of the examinations and tests specified in Section 4. The sanders shall be complete so that when connected to the specified source of power, they can be used for any operation for which they were designed.

3.5.1 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or permanently deformed. All threaded parts shall be in the inch system and shall conform to Handbook H-28.

3.5.2 Gears and shafts. Gears and shafts shall be manufactured from heat treated alloy steel. Gears and shafts shall be supported by at least two bearings and shall be of such dimensions to prevent deflection. Gears shall have machine cut or roll-formed teeth and be designed to efficiently transmit imposed loads. Gears and shafts, except end of spindle shaft, shall be fully enclosed.

3.5.3 Bearings. All rotating parts of the sanders shall be provided with precision quality commercially available ball, roller or needle roller bearings conforming to AFBMA Standards, Section 1 for Terminology, Section 2 for Boundary Dimensions, Section 3 for Tolerance Limits and Section 7 for Mounting Instructions. Bearings shall be fitted according to the bearing Manufacturers recommended fitting practices for thin shell (drawn cup) type needle roller bearings. For bearing identification AFBMA Section 5 or MIL-STD-102 shall apply. Integral or separate shielding to prevent leakage of lubricant and entrance of contaminants shall be provided where necessary. Steel inserts may be omitted in mounting of bearings when sufficient contact area and suitable fit between bearing outer-diameter and housing is provided.

3.5.3.1 Armature shaft bearings. Armature shaft bearings shall be ball, roller, needle roller or a combination thereof providing hardness of shaft, provisions for thrust loading and sealing or shielding are adequate.

3.5.3.2 Spindle bearings. Spindle arbor bearings shall be either ball, roller, taper roller, or needle roller. If roller or needle roller bearings are used, axial thrust loads applied to the spindle shaft must be provided for by use of a separate thrust bearings, either ball or roller, or an integral combination type of roller or needle roller thrust bearing.

3.5.3.3 Thrust provision. Bearings designed primarily for radial loads shall not be used to take thrust. Provision shall be made to absorb thrust loads.

3.6 Housing. Housings shall totally insulate the tool operator from the current carrying parts of the tool. Double-insulated sanders shall have housings of dielectric material or a combination of metal and dielectric material, except Type III, Style 1, Sanders shall have housings of all dielectric material. Dielectric housing material shall

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be of uniform thickness ±1/32 inch of manufacturer's design thickness. Internal metallic skeleton framework or shroud may be incorporated where required to meet the requirements specified herein. All exposed metal parts and attachments shall be completely insulated from current carrying parts and internal metal parts by at least 0.080 inch of protective insulation and by a creepage or clearance distance of at least 3/16 inch or an equal combination of the two. Housings shall include the motor case housing, the spindle housing, when applicable, and handle and switch housing. Color shall be a constituent part of the dielectric material.

3.6.1 Molded parts of housing. Flash shall be removed from molded parts. Flash edges shall be smooth. To prevent excessive moisture absorption into the volume of parts made of thermosetting or thermoplastic material, buffed or cut areas shall be given two coats of varnish conforming to MIL-V-173, if necessary for the sander to meet the 168-hour humidity tests in accordance with 4.5.8.1.

3.7 Electrical equipment. All electrical equipment including motors, switches, and wiring shall conform to applicable requirements of UL 45 in addition to the requirements specified for the following components.

3.7.1 Motor. The motor shall be universal series wound or compensated series designed to operate on 115 volts (+6 percent) single phase 25 to 60 hertz alternating current (AC) or 115 volts direct current (DC). Motors shall conform to the requirements of UL 45.

3.7.1.1 Armature. The armature shall be balanced dynamically to insure freedom from destructive vibration in accordance with commercial practice. Motors shall be constructed to withstand rotational force in excess of their no-load speed.

3.7.1.2 Motor shaft insulation. Reinforced insulation (double insulation) shall be provided between the armature shaft and the lamination core and commutator in accordance with the requirements for commutators and armature end turns of UL 45. Reinforced or protective insulation shall extend axially from under commutator a sufficient distance on both ends of the commutator to prevent a build-up of brush carbon which could result in a deterioration of the shaft-commutator isolation.

3.7.1.3 Commutator. Sander motor commutators may be either of "V" ring and sheet mica or molded construction. The mica shall be undercut between commutator segments. Molded commutators shall be of material that will withstand maximum rotational forces and be immune to deformation from temperatures encountered in service.

3.7.1.4 Brushes. The motor shall be furnished with brushes fitted in suitable insulated brush holders. Brushes not fitted with flexible copper shunts and springs shall be provided with springs made of suitable alloys or bimetallic materials capable of carrying the maximum rated motor current without overheating or degradation of spring characteristics. The minimum internal creepage shall be 3/16 inch, measured from the edge of the live brush holder over the insulation to the adjacent metal part which can be touched from the outside of the housing. Where insulated screw caps are used, they shall be protected by the housing from mechanical injury. Brushes and brush holders shall be such that:

(a) Brush will not rotate in holder.
(b) No live parts are exposed.
(c) Adequate brush pressure is maintained throughout life of brush.
(d) Brush springs or brush springs and shunts shall not be affected by operating temperatures.
(e) Brushes can be easily replaced without deforming any part of the brush holder.

3.7.2 Wiring, cord, and plug. All wiring, flexible cord, and attachment plug caps shall be in accordance with UL 45. The attachment plug cap shall comply with UL 498 requirements for two-pole, two wire devices rated at 15 amperes, 125 volts. Unless otherwise specified (see 6.2), the power cord shall be not less than nine feet long measured from the point of entry to the sander.

3.7.3 Switch. Type I and type II sanders shall have an AC/DC switch of the momentary contact, automatic release, quick break design for turning the motor "on" and "off". Type III sanders shall have an AC/DC switch of the single throw, quick break design and shall have a locking device for holding in the "on" position. All switches shall be encased in high-dielectric insulating material and shall conform to the requirements of UL 45 for switches used in double insulated tools. The switch shall have a current rating of not less than the rated load rating of the sander and shall be capable of interrupting stalled motor current at rated voltage for 50 consecutive contacts at 10-second intervals without sticking or burning of the contacts.

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3.7.4 Electromagnetic interference suppression. When specified (see 6.2), sanders shall be electromagnetic interference suppressed within the limits specified for portable electric equipment in compliance with the requirements of MIL-STD-1337. Suppression components shall not be electrically connected between electrified internal parts and exposed external metal parts.

3.8 Fungus-proofing. When specified (see 6.2), the electrical circuitry, including all components and connections, shall be protected from the effect of water and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, with the following exceptions:

(a) Components or circuit elements which are inherently fungus and water resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

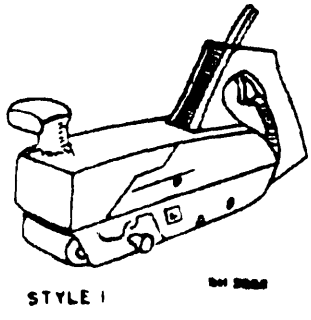
When used, the varnish shall be applied to give a minimum dry film thickness of 0.001 inch to component or element surface previously cleaned and prepared in accordance with accepted commercial practice.

3.9 Type I, belt sander. Belt sanders shall be well balanced and designed for working next to uprights or walls. The belt shall be easy to replace and proper belt tensions shall be maintained under all working conditions. Type I belt sanders shall be similar to figures 1, 2, or 3 and shall conform to Table II for the style specified (see 6.2).

3.9.1 Style 1. Style 1 belt sanders shall be without a dust collecting system (see figure 1).

3.9.2 Style 2. Style 2 belt sanders shall be equipped with an attached dust collecting system (see figure 2).

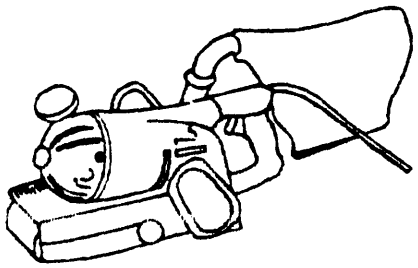
3.9.3 Style 3. Style 3 belt sanders shall be equipped with a collector for attachment to an external dust collecting system (see figure 3).



STYLE I

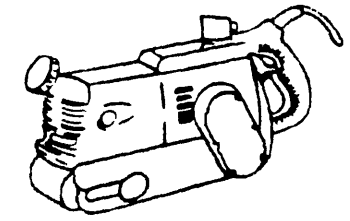
Figure 1 - Type I, style 1, belt sander without dust collecting system.

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STYLE 2

Figure 2 - Type 1, belt sander, style 2, with an attached dust collecting system.



STYLE 3

Figure 3 - Type I, style 3, belt sander with collector for attachment to an external dust collecting system.

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Table II - Type I, belt sanders, styles 1, 2, and 3.

Nominal size	Belt dimensions and speed			Sander dimensions				Net Weight of Tool	
	Width	Length	No-Load	Width	Height	Length	Rated-Load	Without	With
			Speed[1/] range		Without bag (Max.)	Without bag (Max.)	Current at 115 volts A.C. (Min.)	Dust collector (Max.)	Dust collector (Max.)
	(Max.)	(Min.)		(Max.)					
	Inches	Inches	Surface ft. per minute	Inches	Inches	Inches	Amperes[1/]	Pounds	Pounds
1	3	21	800-1300	6	9	13	6	10.5	11.5
2	3	24	900-1600	6	10	18	6	12	15
3	4	27	900-1700	6-3/8	10	18	7	23	25

[1/] Speed at rated load shall be not less than 55 percent of no-load speed.

3.9.4 Components. Belt sanders covered by this specification shall consist essentially of; housing, electric motor, switch, wiring, gears, shafts, bearings, drive mechanism, belt tracking, tensioning mechanisms, sanding shoe, striker bar, handles and accessories.

3.9.4.1 Drive mechanism. The drive mechanism shall transmit the rotary motion of the motor to the sander. Drive mechanism shall be positive drive belt and gear or chain and gear driving a system of pulleys.

3.9.4.2 Belt tracking and tensioning mechanism. Belt tracking and tensioning mechanism shall be conveniently located to insure positive alignment and proper tension of the belt and facilitate easy replacement and adjustments.

3.9.4.3 Sanding shoe. Sanding shoe shall have a resilient pad under a wearing surface of flexible sheet material.

3.9.4.4 Striker bar. Belt sanders shall be equipped with a striker bar designed to prevent belt damage to housing.

3.9.4.5 Handles. Sanders shall have a full hand grip handle on the rear, and a single knob or grip on the front. Handles shall be conveniently located for comfortable operating control of the tool. The operating switch shall be contained in the handle.

3.9.4.6 Standard equipment. Belt sanders shall be complete with abrasive belts and extra parts as normally furnished commercially.

3.9.4.7 Dust-collecting system. Type 1, style 2, sanders shall be equipped with a dust-collecting system that consists of a fan, a vacuum intake duct, and a bag or container into which the dust shall be deposited. The dust bag or container shall be so arranged that the accumulated dust may be conveniently removed. At the option of the contractor, the dust collector may have a separate motor which shall be operated by the same switch as the sander. Type I, style 3 sanders shall be provided with a means of connecting the sanders to an external dust collecting system.

3.10 Type II, disk sander. Disk sanders shall be similar to figures 4 and 5 and shall conform to table III for the type and size specified (see 6.2). Sanders shall be heavy duty types and shall comply with the capacities and dimensions of Table III.

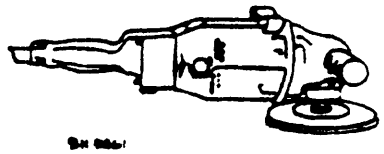
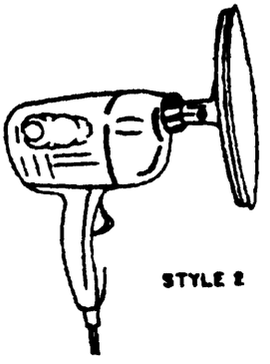


Figure 4 - Type II, style 1, disk sander, straight handle.

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Figure 5 - Type II, style 2, disk sander, side handle.

Table III - Disk, sanders, type II, styles 1 and 2, sizes and capacities.

Requirements	Style 1		Style 2
	Size 7	Size 9	
Abrasive disk diameter, inches	7	9	7
Backing pad diameter, inches, (min.)	6-1/2	8-1/2	6-1/2
Spindle diameter, inches	5/8	5/8	5/8
spindle thread	5/8-11 UNC	5/8-11 UNC	5/8-11 UNC
Spindle speed no load RPM (max.)	5500	5000	5000
RPM at rated horsepower (min.)	1/	1/	1/
Horsepower at spindle (min.)	.70	.70	0.35
Length overall, without backing pad, inches (max.)	20	21	12
Weight, without cable and backing pad, pounds (max.)	13	20	10

1/ Not less than 55 percent of actual no-load speed.

3.10.1 Components. Disk sanders covered by this specification shall consist essentially of housings, electric motor, switch, wiring, gears, spindle shaft, bearings, handles and accessories.

3.10.1.1 Housings, electrical components, gears and bearings shall conform to the requirements specified herein.

3.10.1.2 Spindle. Spindle shall be hardened alloy or forged steel and shall be supported by at least two bearings. Outer spindle bearing shall be ball or roller or a combination ball and roller designed to withstand the radial thrust loads imposed. Spindle thread size shall conform to table III and ANSI B5.38. Means shall be provided for locking or holding spindle to facilitate changing abrasive disks and backing pads.

3.10.1.3 Handle. Style 1 disk sanders shall have a straight type handle at the end opposite the sanding disk. Style 2 disk sanders shall have a side handle. Handles shall house the switch and be designed to provide a convenient grip for the operator. Sanders shall have an auxiliary side handle provided for attachment to either the front, right or left side of the spindle or motor housing.

3.10.1.4 Standard equipment backing pad. Disk sanders shall be furnished with a replacement semi-flexible backing pad for the abrasive disk. Pads shall be constructed of a material which will not deteriorate from the heat created by sanding and shall withstand the maximum centrifugal force. Pad shall have a recessed center and shall be of such size and thickness as to provide maximum support to the sanding disk. Backing pad shall conform to Table III for the size sander specified.

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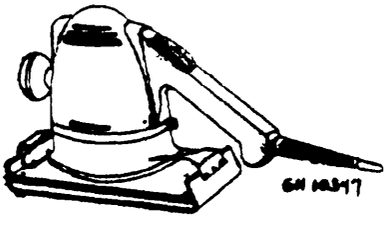
3.10.1.5 Tool rest. Type II, style 1, disk sanders shall be equipped with tool rest of suitable resilient material fastened to the housing opposite the sanding disk to enable resting the tool without injury to the sander spindle and disk assembly. When specified (see 6.2), style 1 disk sanders shall have provision for suspension eyebolt mounting.

3.11 Type III, sanders orbital pad. Orbital pad sanders shall be similar to figures 6 and 7 and shall conform to table IV for type and style specified (see 6.2).



ONE HAND OPERATION

Figure 6 - Type III, style 1, orbital pad sander, one hand grip.



TWO HAND OPERATION

Figure 7 - Type III, style 2, orbital pad sander, two hand grip.

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Table IV - Type III, orbital pad sanders, styles 1 and 2.

Style	Abrasive area (Min.)	Amplitude of Oscillation (Min.)	Amplitude of Oscillation (Max.)	Weight less cable (Max.)	Test weight	Work factor (Min.)
	Square inches	Inch	Inch	Net-Pounds	Pounds	
1	18	.078	.250	3-3/4	10	225
2	25	.078	.250	6-1/2	10	300

3.11.1 Components. Orbital sanders covered by this specification shall consist essentially of housings, electric motor, switch, wiring, motion transfer mechanism, backing pad and abrasive holding device, handles and accessories.

3.11.1.1 Housings, electrical components, gears, shafts, and bearings shall conform to the requirements specified herein.

3.11.1.2 Motion transfer mechanism. Sanders shall have transfer mechanism to transfer motor rotation to an orbital motion of the sanding pad. Moving elements of the mechanism shall be mounted in ball bearings designed to withstand the imposed radial thrust loads. Motive power shall be transmitted to the pad by belt, cam, gear train, direct drive or other means.

3.11.1.3 Abrasive holding device. Each sander shall be furnished with an abrasive holding device designed to quickly remove and securely replace abrasive paper without the use of special tools. If non-standard tools are required to replace abrasive paper, they shall be permanently attached to the sander and readily available for use.

3.11.1.4 Handles. Type III, style 1 sanders, designed for one-hand operation shall have a functional single grip. Style 2, sanders, shall have a side handle, top-mounted or side-mounted on the motor housing and an auxiliary handle.

3.11.1.5 Standard equipment. Type III sanders shall be equipped with a backing pad of felt, rubber or other durable resilient material attached to a metal plate.

3.12 Identification plate. A substantial corrosion resistant identification plate, permanently and legibly marked to include the following information shall be securely attached to the sander or as an integral part of the housing-also the words "Double Insulation," or symbol denoting double-insulation shall be included on the plate or in a prominent position on the sander.

(a) Manufacturer's name.
(b) Manufacturer's model or serial number.
(c) Voltage.
(d) Rated-load, current, amperes (at 60 hertz).
(e) No-load surface feet per minute (sfpm) for Type I, no-load RPM for Type II Sanders; and (orbits per min.) O.P.M. for type III Sanders.

3.13 Handbook of instruction. Unless otherwise specified (see 6.2), the manufacturer shall furnish one copy of his regular instruction book and parts lists with each sander. The book shall include all illustrations, instructions, and necessary data for safe operation, maintenance, adjustment, and repair. The parts list shall include the manufacturer's part number and ordering instructions. Sanders shall have one of the following notations on the identification plate.

"This tool requires two (2)-prong plug only."

"Grounding not required."

3.14 Tools and extra parts. In addition to equipment specified herein, each sander shall be furnished with such maintenance tools, and extra parts as normally furnished commercially. Additional accessory equipment shall be as specified (see 6.2).

3.15 Workmanship. Workmanship of the sander and accessories shall be of the quality prevailing among manufacturers normally producing sanders of the double-insulated type specified herein and shall be first class in every respect.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of all the examinations of 4.4.1 and tests in 4.4.2 required for quality conformance and the additional tests of 4.5.5 through 4.5.8.8. Failure of the sander in any tests shall result in the termination of the qualification tests. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.5 and 6.6).

4.2.1 Prior approval of electromagnetic interference suppression. Prior to authorization for qualification testing, complete electromagnetic interference tests as specified herein shall be performed by the contractor or a commercial test laboratory to determine compliance with the interference suppression requirements. Three copies of the interference suppression test report shall be forwarded to the Naval Ship Engineering Center for review and approval.

4.2.2 Samples for qualification tests. For type I, the manufacturer shall submit one style 2, size 2 belt sander. For type II, the manufacturer shall submit one style 1, 7-inch and one style 1, 9-inch disk sander. For type III, the manufacturer shall submit one style 1 and one style 2 orbital pad sander. Approval of the type I, style 2, size 2 will extend approval to type I, styles 1 and 3, sizes 1 and 3. Approval of type II, style 1, sizes 7 and 9 will extend approval to type II, style 2. Sanders submitted for qualification tests shall be complete with electromagnetic interference suppression system installed, and with instruction book, parts list, and two extra sets of brushes.

4.3 Sampling for quality conformance.

4.3.1 Lot. All sanders of the same type, style, and size offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

4.3.2 Sample. A random sample shall be selected from each lot in conformance with MIL-STD-105 at inspection level II.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on each sample sander selected in accordance with 4.3.2. Quality conformance inspection shall consist of the following:

(a) Examination 4.4.1.
(b) Tests 4.4.2.
(c) Inspection of preservation and packaging 4.6.

4.4.1 Examination. Each sander selected in accordance with 4.3.2 shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with Table V, acceptable quality level (AQL) to be as indicated. Nonconforming sanders shall not be offered for delivery and if the nonconforming sanders in any one sample exceed the acceptance number for that sample, the lot represented shall be rejected.

Table V - Classification of defects.

Categories	Defects
Critical:	AQL 1.0 percent defective.
001	Moving parts (hazardous to operator) not adequately guarded with guards which do not interfere with normal operation of sander.
Major:	AQL 2.5 percent defective
101	Size not as specified.
102	Handles not as required
103	Cord and plug not as specified.

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Table V - Classification of defects (continued).

Categories	Defects
Major (Cont'd):	
104	Locking device for switch not included.
105	Intake air filter missing.
106	Type I Sanders, dust-collecting system not as specified.
107	Type II Sanders, spindle size and thread not as specified.
108	Type III Sanders, not one-hand or two-hand operation as specified.
109	Standard equipment or extra parts or accessories not as specified.
110	Workmanship not of quality expected of portable electric commercial equipment.
Minor:	AQL 4.0 percent defective.
201	Handbook of instructions not as specified.
202	Identification plate not as specified.

4.4.2 Tests for quality conformance. Each sample sander selected in accordance with 4.3.2 shall be tested in compliance with the requirements of this specification. AQL shall be 2.5 percent defective. Nonconforming sanders shall not be offered for delivery and if the number of defective units exceeds the acceptable number for the sanders for the samples tested, the lot represented shall be rejected. Quality conformance tests shall consist of the tests specified in 4.5.3, and 4.5.4.

4.5 Tests.

4.5.1 Test procedures shall be as follows:

(a) Temperature - All performance tests shall be conducted in an ambient temperature of 77° ±20°F.
(b) All voltages are RMS values unless otherwise stated.
(c) Humidity tests of 4.5.8.1 and 4.5.8.4 shall be prepared using a saturated solution of Potassium Sulphate (K_2SO_4) dissolved in distilled water. Undissolved crystals shall be visible in solution container at all times. 95 ±3 percent R.H. shall be controlled by maintaining cabinet temperature between 20 - 30 degrees centigrade.
(d) The electromagnetic interference suppression system shall be disconnected during the tests of 4.5.8.5 and 4.5.8.7.

4.5.2 Electromagnetic interference test report. When specified (see 6.2), manufacturer shall provide certification that the electromagnetic suppression network in the sander offered is identical to that of sanders previously tested and approved prior to qualification.

4.5.3 Dielectric strength electrical circuits. Each sample sander selected shall be tested for electric circuit dielectric strength by withstanding 4000 volts RMS, 60 cycle, alternating current from line to exposed metal surface with switch "on" for a period of one second.

4.5.4 Operational tests.

4.5.4.1 No-load test. Sample sanders selected for test shall be operated at no-load for not less than 30 minutes.

4.5.4.2 Rated-load test. Upon completion of the no-load test, the sander shall be operated at rated-load for not less than 20 minutes. Voltage frequency and current shall be in accordance with the sander nameplate data. Load shall be in accordance with the loads for tests requirements of UL 45. During the test the sanders shall be checked for switch operation and loss of lubricant. Leakage current shall be determined and shall not exceed one milliampere. Abrasive belt tracking and drive mechanism tension shall be examined for faulty operation when applicable. Evidence of off balance or expansion of backing pad on disk sanders shall be determined.

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4.5.5 Performance tests. Sample sanders shall be tested for the requirements specified herein and as listed on the identification plate:

Type I Sanders - (a) Belt speed (Surface Feet Per Minute)
(1) No-load
(2) Rated-load

Type II Sanders - (a) Horsepower output at spindle as specified in table III.
(b) Spindle speed - (rpm)
(1) No-load
(2) Rated-load in accordance with table III.

Type III Sanders - (a) Speed - (Orbits Per Minute)
(1) No-load
(2) Rated-load

Type I, II, and III Sanders - (a) Rated load current input, at rated voltage and frequency

4.5.6 Work factor test. Type III sanders shall show a work value at test load in compliance with Table IV calculated as follows:

WORK FACTOR - The work factor is a measure of a performance and is calculated by the following formula:

$$\text{Work Factor} = \frac{a\ d\ n\ w}{1{,}728}$$

Where: a = abrasive contact area in square inches
d = distance, inches, of travel for one complete cycle of oscillation. For reciprocating motion sanders, the distance will be the amplitude times 2. For orbital motion sanders, the distance will be the perimeter of the orbit.
n = number of oscillations per minute, using the test weight as specified in Table IV and using the following abrasive against a steel surface: No. 60 grit aluminum oxide, closed coating, medium weight cloth backing.
w = test weight

NOTE: See 6.6 for definition of abrasive contact area, oscillations and test weight.

4.5.7 Motor tests. Each sample sander motor shall be tested for temperature rise under rated load. The motor shall be run at rated load in accordance with the nameplate data until temperature has stabilized, but not less than 1/2 hour. The motor winding temperature shall not exceed 65°C. above ambient temperature when measured by thermocouple method.

4.5.8 Protective insulation tests. The protective insulation tests shall be conducted in the following order: 4.5.8.1, 4.5.8.2, 4.5.8.3, 4.5.8.4, 4.5.8.5, 4.5.8.6, 4.5.8.7, and 4.5.8.8. The inside of the sander housing and the sander internals shall not be cleaned in any way during this test sequence.

4.5.8.1 Humidity test. Before any other tests are run, the sander shall be placed in a humidity chamber for a period of one week (168 hours) at 95 percent relative humidity and 25° to 30°C. At the end of the humidification period and within 5 minutes after removal from the humidity chamber, the insulation resistance and leakage current shall be measured and recorded. The insulation resistance shall not be less than 1 megohm and the leakage current shall not exceed 3.5 milliamperes.

4.5.8.2 Sander no-load running and brush wear test. The sander shall be run at no-load (with normal air flow through sander for 100 hours or until the brushes wear out. Failure of any components during this test shall be cause for rejection of the sander.

4.5.8.3 Leakage current. When the test of 4.5.8.2 is concluded, leakage current shall be measured from each power supply conductor to accessible metal parts and to metal foil applied to the exterior surface of the protective insulated housing. The leakage current shall not exceed 1 milliampere (0.001 amperes). Applied voltage shall be 110 percent of the rated voltage marked on the nameplate. A new set of motor brushes shall be installed before any further tests are conducted.

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4.5.8.4 Insulation resistance test. Sander shall be placed in a humidity cabinet for 48 hours. Temperatures in the cabinet shall be 20° to 30°C. and a relative humidity of between 90 and 95 percent. Upon removal from the cabinet the insulation resistance shall be measured using an applied potential of 500 volts DC between live parts via plug and switch and accessible dead metal parts and between live parts and metal foil applied to the exterior surface of the protective insulated housing or case. Minimum insulation resistance shall be not less than 1 megohm. Nonconforming sanders shall be rejected.

4.5.8.5 Insulation dielectric tests. A dielectric test shall be performed immediately after the insulation resistance test. Sixty hertz potential shall be applied for at least one minute without breakdown at the following voltage:

(a) Between live metal parts and inaccessible dead metal parts 1000 volts plus 2 times rated voltage.
(b) Between inaccessible dead metal parts and accessible dead metal parts and also between inaccessible dead metal parts and metal foil applied to exterior surface of sander 2,000 volts plus 2 times rated voltage.
(c) Between metal foil applied to interior of compartments containing internal wiring and accessible dead metal parts or metal foil attached to exterior surface of the sander 2,000 volts plus 2 times rated voltage.
(d) Between metal foil wrapped around power cord at the junction with tool and metal foil attached to exterior of sander or accessible dead metal parts 2000 volts plus 2 times rated voltage.
(e) Between live parts and accessible dead parts and between live parts and metal foil applied to exterior of sander 3,500 volts plus 2 times rated voltage.

4.5.8.6 Impact tests. The sander shall be tested as follows for compliance with the impact resistance qualities of the protective insulating material: The sander shall be dropped (without wheel) at least 3 times on a hard wood surface from a distance of 3 feet in such a position that the sander will strike in a different position each drop. After the above tests, the sander shall be examined and will be considered as having failed the tests if there is:

(a) Evidence of cracking case or housing.
(b) Reduction of spacing below minimum required.
(c) Evidence that internal parts have been damaged or sander fails to operate due to broken insulation or internal parts.

4.5.8.7 Insulation stability test. (This is a destructive test to be conducted for the purpose of determining safety properties of insulation.) The sander shall be connected to rated voltage frequency and loaded using a dynamometer, prony brake or other loading device. The current input shall be 1.8 times the rated input marked on the tool. This load shall be maintained for one hour or until the motor fails. Following this overload, the following requirements shall be met:

(a) There shall be no exposed live parts.
(b) Spacings shall not be less than the minimum required by this specification.
(c) A potential of 2,500 volts 60 hertz shall be applied for one minute without breakdown between live parts and accessible dead metal parts and between live parts and metal foil applied to the exterior surface of housing made of insulating material.

4.5.8.8 Flammability test. At least three sample specimens 1/2 inch wide, 5 inches long by the material thickness, shall be cut from a sheet or molded of the identical protective insulation used in fabrication of the tool or from tool test sample housings. Sample specimens shall conform to the requirements and test procedures of method 2021 of Federal Test Method Standard No. 406.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with Table VI. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable level (AQL) of 4.0 defects per hundred units.

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Table VI - Classification of preparation for delivery defects.

Examine	Defects
Markings (exterior and interior)	Omitted: incorrect; illegible; improper size, location, sequency, or method of application.
Materials	Any components missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strappings, or distortion of container.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.7.)

5.1 Preservation, packaging, packing, and marking. Sanders shall be preserved, packaged, packed, and marked in accordance with PPP-P-40. The level of preservation and packaging, and the level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. Type I sanders are intended for fast, heavy sanding and straight-line finishing of wood and other materials. Type II sanders are intended for smoothing surfaces, buffing and polishing of metal and other materials. Type III sanders are intended for finish sanding of wood and other materials. Double insulated sanders are particularly recommended for use aboard ships or in damp locations and outside work because of their shockproof design.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, style, and size required (see 1.2).
(c) Specify cord length required if different (see 3.7.2).
(d) Electromagnetic interference suppression when required (see 3.7.4 and 4.5.2).
(e) Moisture and fungus resistance treatment when required (see 3.8).
(f) Eyebolt mounting when required (see 3.10.1.5).
(g) Specify handbook of instruction if different (see 3.13).
(h) Specify additional accessory equipment required (see 3.14).
(i) Level of preservation and packaging, and packing required (see 5.1).

6.3 Reclassification. The Size 7 and Size 9 disk sanders have been reclassified as Type II disc sanders, Size 7 and Size 9.

6.4 Supersession data. This specification includes the requirements of Military specification MIL-S-24115A(SHIPS) dated 9 September 1965.

6.5 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 00111C, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.5.1).

6.5.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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6.6 Definitions. Definitions are as follows:

6.6.1 Double-insulation. Insulation comprising both functional insulation and protecting insulation, with the two insulations physically separated and so arranged that they are not simultaneously subjected to the same deteriorating influences (temperatures and contaminants) to the same degree.

6.6.1.1 Functional insulation. Insulation necessary for the proper functioning of the tool and for basic protection against electrical shock.

6.6.1.2 Protecting insulation. Denotes an independent insulation provided, in addition to the functional insulation. An enclosure of insulating material may form a part or the whole of the protecting insulation.

6.6.1.3 Reinforced insulation. Denotes an improved functional insulation with such mechanical and electrical qualities that it provides the same degree of protection against electrical shock as double-insulation. The enclosures of insulating material may form a part or the whole of the protecting insulation.

6.6.2 Abrasive contact area. The area of actual contact between the abrasive and the work when the machine is at rest with the test weight applied.

6.6.3 Oscillations. The number of complete cycles per minute when the machine is operating with the test weight applied. In case of orbital motion sanders, a cycle is the describing of one complete orbit. In case of reciprocating sanders, a cycle is one complete back and forth movement.

6.6.4 Amplitude. The distance of travel in one direction for reciprocating motion sanders; and the maximum diameter of the orbit motion sanders.

6.6.5 Test weight. The test weight shall consist of the actual weight of any particular sander less cable, plus an extra weight added to the top or handles of the sander to bring the total up to the test weight indicated in Table III. For example, a 7-pound sander would have 7 pounds added in order to build up to the test weight.

6.7 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "*" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity as indicated in reverse hereof.

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

FORM 1426
DEC 65

REPLACES DD FORM 1426, WHICH IS OBSOLETE

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